

Subnetting 101: identifying the number of Subnets

CIDR notation – a short hand for subnets

/24 is for our standard subnet with 255 hosts (255.255.255.0) (11111111 11111111 11111111 00000000)

(Note the first 3 binary octets are the network portion of the address and the final octet (full of zeros) is the host portion)

The 24 designation is for the 24 1's that make up the network portion)

A subnet mask will always have consistent 1's (network portion) and consistent 0's (host portion)

/20 subnet /20 is 20 1's followed by remaining 12 0's) (Four Octets equals 32 digits or 4 to 8 binary digits)

/20 subnet in binary is 1111|1111 1111|1111 1111|0000 0000|0000

Which converts to 255.255.240.0

Formulas

Hosts -> 2^{n-2}

Subnets -> 2^n

1. When asked for the number of **hosts**, use the formula: Host = 2 to the power of n -2

(Note: Count the Host Bits from Right to Left Networks = 2 to the power of n or 2^n)

2. When asked for the number of **subnets**, use the formula. Networks = 2^N (N being the factor that is equal to or more than our desired amount of subnets) (example make 20 subnets 2^4 is 16 and 2^5 is 32, thus 16 is not enough and we will need to take 5 bits from the host section and add them to the network section, leaving us with 32 available subnets)

/25 /26 /27 /28 /29 /30 /31 /32 - 4th

128 192 224 240 248 252 254 255 = mask

128 64 32 16 8 4 2 1 = increment

1 2 3 4 5 6 7 8 = bits

Subnets Formula

Example 1: Let's say we want 16 subnets for our network

$16 = 2^4$ (Therefore $N=4$)

Thus steal 4 bits from hosts for and give them to network portion. (represent the subnet portion in Green)

10.1.1.0/24

255.255.255.(0000|0000) take 4 bits from the hosts leaving you with **255.255.255.(0000|0000)** (Green portion is subnets / Blue is Hosts for these subnets)

	Class A	Class B	Class C
1 st Octet Range	1 – 126	128 – 191	192 – 223
Network Mask	255.0.0.0	255.255.0.0	255.255.255.0
2 Rules			

Example Question 1: How many subnets for this /12 network?

How many valid subnets exists on 10.0.0.0 subnet 255.240.0.0 or in CIDR notation (10.0.0.0/12)

255.240.0.0 in binary is (1111|1111 1111|0000 0000|0000 0000|0000) (Since there are 12 consecutive 1's it is /12 in cidr)

10.0.0.0 in binary is (0000|1010 0000|0000 0000|0000 0000|0000) (Red is Network | Blue is Host)

Subnet Formula = 2^N

12

CIDR Conversion Table

CIDR prefix length	Dotted Decimal Netmask	Hexidecimal Netmask	Inverse Netmask	Binary	Number of Classfull Networks	Number of Usable IPs
/1	128.0.0.0	80 00 00 00	127.255.255.255	1000 0000 0000 0000 0000 0000 0000 0000	128 A's	2,147,483,648
/2	192.0.0.0	C0 00 00 00	63.255.255.255	1100 0000 0000 0000 0000 0000 0000 0000	64 A's	1,073,741,822
/3	224.0.0.0	E0 00 00 00	31.255.255.255	1110 0000 0000 0000 0000 0000 0000 0000	32 A's	536,870,910
/4	240.0.0.0	F0 00 00 00	15.255.255.255	1111 0000 0000 0000 0000 0000 0000 0000	16 A's	268,435,454
/5	248.0.0.0	F8 00 00 00	7.255.255.255	1111 1000 0000 0000 0000 0000 0000 0000	8 A's	134,217,726
/6	252.0.0.0	FC 00 00 00	3.255.255.255	1111 1100 0000 0000 0000 0000 0000 0000	4 A's	67,108,862
/7	254.0.0.0	FE 00 00 00	1.255.255.255	1111 1110 0000 0000 0000 0000 0000 0000	2 A's	33,554,430
/8	255.0.0.0	FF 00 00 00	0.255.255.255	1111 1111 0000 0000 0000 0000 0000 0000	1 A or 256 B's	16,777,214
/9	255.128.0.0	FF 80 00 00	0.127.255.255	1111 1111 1000 0000 0000 0000 0000 0000	128 B's	8,388,606
/10	255.192.0.0	FF C0 00 00	0.63.255.255	1111 1111 1100 0000 0000 0000 0000 0000	64 B's	4,194,302
/11	255.224.0.0	FF E0 00 00	0.31.255.255	1111 1111 1110 0000 0000 0000 0000 0000	32 B's	2,097,150
/12	255.240.0.0	FF F0 00 00	0.15.255.255	1111 1111 1111 0000 0000 0000 0000 0000	16 B's	1,048,574
/13	255.248.0.0	FF F8 00 00	0.7.255.255	1111 1111 1111 1000 0000 0000 0000 0000	8 B's	524,286
/14	255.252.0.0	FF FC 00 00	0.3.255.255	1111 1111 1111 1100 0000 0000 0000 0000	4 B's	262,142
/15	255.254.0.0	FF FE 00 00	0.1.255.255	1111 1111 1111 1110 0000 0000 0000 0000	2 B's	131,070
/16	255.255.0.0	FF FF 00 00	0.0.255.255	1111 1111 1111 1111 0000 0000 0000 0000	1 B or 256 C's	65,534
/17	255.255.128.0	FF FF 80 00	0.0.127.255	1111 1111 1111 1111 1000 0000 0000 0000	128 C's	32,766
/18	255.255.192.0	FF FF C0 00	0.0.63.255	1111 1111 1111 1111 1100 0000 0000 0000	64 C's	16,382
/19	255.255.224.0	FF FF E0 00	0.0.31.255	1111 1111 1111 1111 1110 0000 0000 0000	32 C's	8,190
/20	255.255.240.0	FF FF F0 00	0.0.15.255	1111 1111 1111 1111 1111 0000 0000 0000	16 C's	4,094
/21	255.255.248.0	FF FF F8 00	0.0.7.255	1111 1111 1111 1111 1111 1000 0000 0000	8 C's	2,046
/22	255.255.252.0	FF FF FC 00	0.0.3.255	1111 1111 1111 1111 1111 1100 0000 0000	4 C's	1,022
/23	255.255.254.0	FF FF FE 00	0.0.1.255	1111 1111 1111 1111 1111 1110 0000 0000	2 C's	510
/24	255.255.255.0	FF FF FF 00	0.0.0.255	1111 1111 1111 1111 1111 1111 0000 0000	1 C	254
/25	255.255.255.128	FF FF FF 80	0.0.0.127	1111 1111 1111 1111 1111 1111 1000 0000	1/2 C	128
/26	255.255.255.192	FF FF FF C0	0.0.0.63	1111 1111 1111 1111 1111 1111 1100 0000	1/4 C	62
/27	255.255.255.224	FF FF FF E0	0.0.0.31	1111 1111 1111 1111 1111 1111 1110 0000	1/8 C	30
/28	255.255.255.240	FF FF FF F0	0.0.0.15	1111 1111 1111 1111 1111 1111 1111 0000	1/16 C	14
/29	255.255.255.248	FF FF FF F8	0.0.0.7	1111 1111 1111 1111 1111 1111 1111 1000	1/32 C	6
/30	255.255.255.252	FF FF FF FC	0.0.0.3	1111 1111 1111 1111 1111 1111 1111 1100	1/64 C	2
/31	255.255.255.254	FF FF FF FE	0.0.0.1	1111 1111 1111 1111 1111 1111 1111 1110	1/128 C	0
/32	255.255.255.255	FF FF FF FF	0.0.0.0	1111 1111 1111 1111 1111 1111 1111 1111	1/256 C	1